Avoiding barren plateaus in a quantum neural network

Track 2 (Quantum Machine Learning), Problem 4 | Qiskit Fall Fest Hackathon, IISER Bhopal | 17 October 2026

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1. Problem and what we set out to test

A deep, randomly initialised quantum circuit can have gradients that shrink exponentially with the number of qubits: the barren plateau. This repository measures that effect, tests a standard mitigation (near-identity initialisation), and then compares the same circuit in three regimes - an exact statevector simulator, the shot-sampled Aer simulator, and one real IBM Quantum device.

Why shots alone are not enough. Training can stall as the gradients become exponentially small. Increasing the number of measurement shots can reduce the sampling uncertainty (the statistical noise in each cost estimate), but it cannot by itself eliminate an exponentially vanishing underlying gradient - the problem is the shape of the loss landscape, not the sampling noise. The measurement protocol below therefore uses exact statevector costs for the gradient study, and reserves Aer shot noise for a separate comparison with hardware.

2. Method

Circuit. A hardware-efficient ansatz: three layers of single-qubit RY rotations on every qubit, each layer followed by a chain of CNOTs between neighbours. The cost is global, C(theta) = 1 - P(|0...0>), which asks every qubit for the answer at once and is the standard way to provoke a plateau.

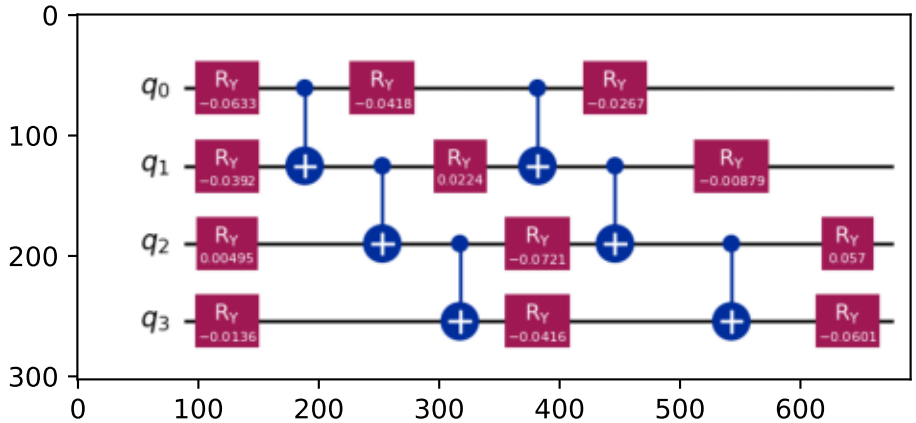


Figure 1. The 4-qubit, 3-layer ansatz as drawn in the original Colab session. The circuit is the same for both initialisations; only the parameter values differ (random: uniform on [0, 2pi), near-identity: uniform on [-0.1, 0.1]).

Gradients. The parameter-shift rule, dC/dtheta = [C(theta + pi/2) - C(theta - pi/2)] / 2, which is exact for an RY angle. It is not an approximation and not a statistical shortcut: every cost here is evaluated on a simulated quantum state.

Measurement protocol. For each width n = 2 ... 10 we draw 100 independent parameter sets, evaluate one gradient component with the parameter-shift rule, and take the sample variance. The standard error of each sample variance is estimated with the normal-theory approximation SE = Var * sqrt(2/(T-1)), where T is the number of trials; this is an approximation (the true distribution of a sample variance depends on higher moments of the gradient), not an exact result, but it is the standard way to attach error bars to a variance estimated from a finite sample. Each set of points is then fitted to an exponential decay model Var ~ A exp(-b n), and the fitted decay rate b is reported with its own uncertainty from the weighted fit. A single run says nothing about a variance; b = 0 means no plateau. Two initialisations are compared: random angles uniform on [0, 2pi) and near-identity angles uniform on [-0.1, 0.1].

Simulators and hardware. Everything is simulated except one hardware run. The gradient-variance study uses the exact noiseless statevector, so the only randomness is in the parameters. The shot-noise study builds the circuit, measures it and counts outcomes on Qiskit Aer - a physical circuit simulation with 1024-shot sampling, not a formula. The same 4-qubit circuit, with the same angles, was submitted once to real IBM hardware. Seeds: 42 for every simulator experiment; the hardware job is a single real run.

3. Results

3.1 The plateau, and the mitigation. With a random initialisation the gradient variance falls by a factor of 25,840 between n = 2 and n = 10, and the fit gives b = 1.256 +/- 0.260 per qubit (Var ~ 5.83e-01 * exp(-b n)). That is an exponential vanishing with width, the textbook signature of a barren plateau. Near-identity initialisation removes it: b = 0.054 +/- 0.287, consistent with zero, and at n = 10 the variance is 506 times larger than with random angles. Because both curves are measured on the same circuit with the same seeded draws, the comparison is direct.

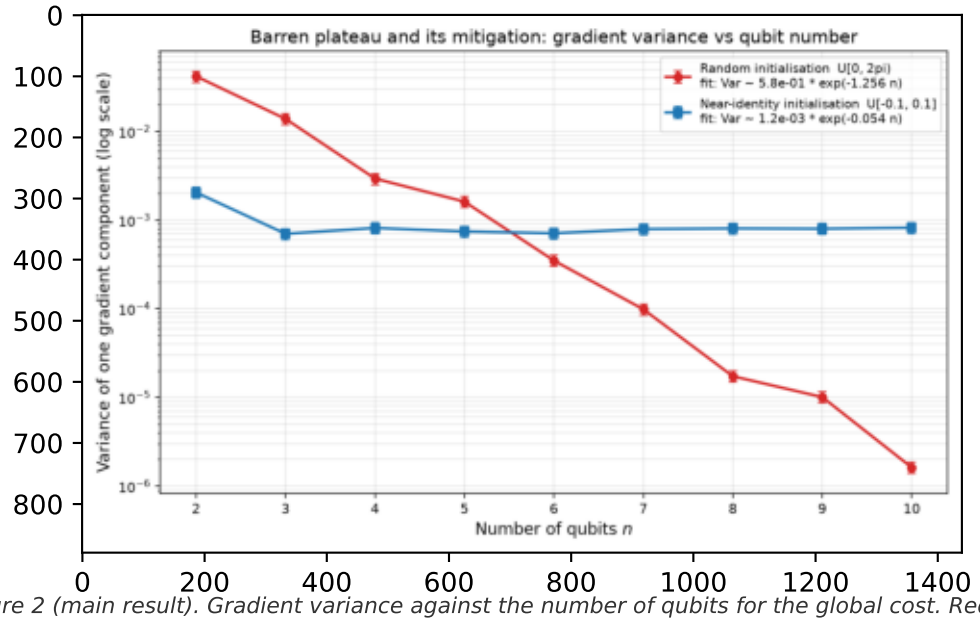


Figure 2 (main result). Gradient variance against the number of qubits for the global cost. Red: random initialisation, exponential decay, the plateau. Blue: near-identity initialisation, no decay. One gradient component, 100 random parameter sets, three-layer ansatz; bars are the standard error of each variance and the legend gives the fitted decay rate per qubit.

3.2 Why it works. A global observable touches every qubit, so its gradient at a scrambled point is a sum of exponentially many cancelling contributions. The near-identity start never visits scrambled points, so its gradients keep a typical size that does not decay with width.

3.3 Shots and real hardware. At 1024 shots the exact cost of the 4-qubit circuit is 0.917173. Across 20 independent Aer runs the mean estimated cost is 0.923193, with a sample standard deviation of 0.007782 across those runs. The expected shot-noise standard deviation for a single cost estimate - the σ of one Bernoulli-sampled cost - is 0.008613, predicted by sqrt(C(1-C)/1024), and the observed mean absolute error relative to the exact cost is 0.007833. These two numbers play different roles: the sample standard deviation describes how much the Aer estimates scatter from run to run, while the expected shot-noise standard deviation is the theoretical σ of a single estimate from binomial sampling. They agree within the usual sampling uncertainty of a variance, so the Aer sampling behaves exactly as it should. The same circuit, with the same angles, was submitted to ibm_fez (job db97h70c77vs738116qg, 1024 shots, 2026-10-17 00:48 IST, transpiled depth 33): the cost came back 0.938477 against 0.917173 in exact simulation and 0.923193 for the mean of the 20 Aer runs. The hardware result is about 2.5 expected shot-noise standard deviations above the ideal value. Gate and readout errors may contribute to this discrepancy, but a single 1,024-shot hardware run cannot separately distinguish hardware errors from sampling variability, and we do not claim that it does.

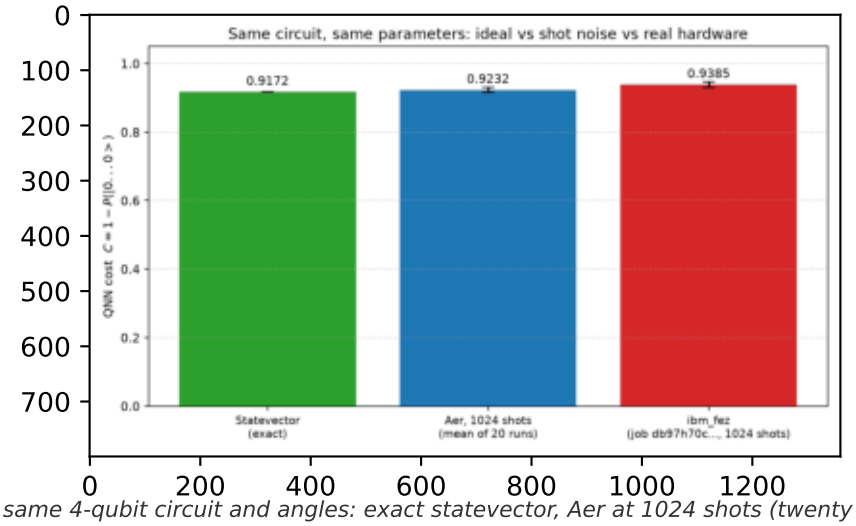


Figure 3. The same 4-qubit circuit and angles: exact statevector, Aer at 1024 shots (twenty runs, bar shows one shot-noise s.d.) and the recorded IBM hardware run on ibm_fez.

4. Honest limits

Widths up to 10 qubits on a statevector simulator are far from the regime where plateaus make algorithms impossible, and every variance here is measured on one gradient component of the first layer; other components and other ansatz layouts may behave differently. The exponential fits have chi-squared per degree of freedom well above one, because the sample variances are noisy, so the decay rates are indicative rather than precise.

Only one hardware job was run, with one set of angles; a noise model was not fitted, so the hardware comparison is qualitative. The mitigation was measured on the gradient variance, which is the quantity the plateau theory predicts; whether an end-to-end training run benefits at larger widths is a separate question we have not tested here.

5. AI use, and what we checked ourselves

AI assistants were used for planning and for refactoring the Colab session into this repository. One concrete thing they got wrong is described in AI_USE_NOTE.md. The repository's tests check the simulator against analytic closed forms, the gradients against central finite differences, and the recorded hardware angles against the exact cost they must reproduce. Every number in this write-up is read from results/*.json at build time by docs/make_writeup.py, so the text cannot drift from the measurements.

References

Cerezo et al., Variational quantum algorithms: a review of methods and the path to quantum advantage, Nature Reviews Physics 3, 625 (2021) - plateaus, global vs local cost functions. McClean et al., Barren plateaus in quantum neural network training landscapes, Nature Communications 9, 4812 (2018). Grant et al., An initialization strategy for addressing barren plateaus in parametrized quantum circuits, Quantum 3, 214 (2019) - identity-block initialisation. Mitarai et al., Quantum circuit learning, Physical Review A 98, 032309 (2018) - parameter-shift rule. Qiskit (2026), Qiskit Aer and IBM Quantum documentation.